Q-2:

Assignment 2

Grade: Check Plus

1.)

x	f(x)	x f(x)	$(x-25.41)^2 f(x)$
20	0.05	1	1.46
21	.06	1.26	1.17
22	0.07	1.54	0.81
23	0.08	1.84	0.46
24	0.09	2.16	0.16
25	.12	3	0.02
26	.15	3.9	0.05
27	0.13	3.51	0.32
28	0.10	2.8	0.67
29	0.10	2.9	1.29
30	0.05	1.5	1.05
		Σ 25.41	Σ 7.46

x Q1: Needs revision.

1.) Mean = 25.41 Variance = 7.46 SD = 2.73

2.) a) $P(x) = -1SD \leq 25.41 \leq +1SD$

$= 22.68 \leq 25.41 \leq 28.14$

$= P(23) + P(24) + P(25) + P(26) + P(27) + P(28)$

✓ Q2a

$= 0.08 + 0.09 + .12 + .15 + .13 + .10$

$= .67$ or $\underline{67\%}$

b) $P(X) =$ Mean $\pm$ 2SD

✓ Q2b

$P(X) = 19.95 \leq 25.41 \leq 30.87$

$P(X) = P(20) + P(21) + P(22) + P(23) + P(24) + P(25) + P(26) + P(27)$
$+ P(28) + P(29) + P(30) = 1.00$ or $\underline{\underline{100\%}}$.

3

✓ Q3

X	F(X)	Value (v)	~~Mean of~~ (v) F(X)	$(v-1.36)^2$ F(x)
20	0.05	0	0	0.09
21	0.06	0	0	0.11
22	0.07	0	0	0.13
23	0.08	0	0	0.15
24	0.09	0	0	0.17
25	.12	0	0	0.22
26	0.15	1	.15	0.02
27	.13	2	.26	0.05
28	.10	3	.30	0.27
29	.10	4	.40	0.70
30	.05	5	.25	0.67

See last page for histogram

4 Option +ve Value if X is 26 or more

✓ Q4

$= P(26) + P(27) + P(28) + P(29) + P(30)$

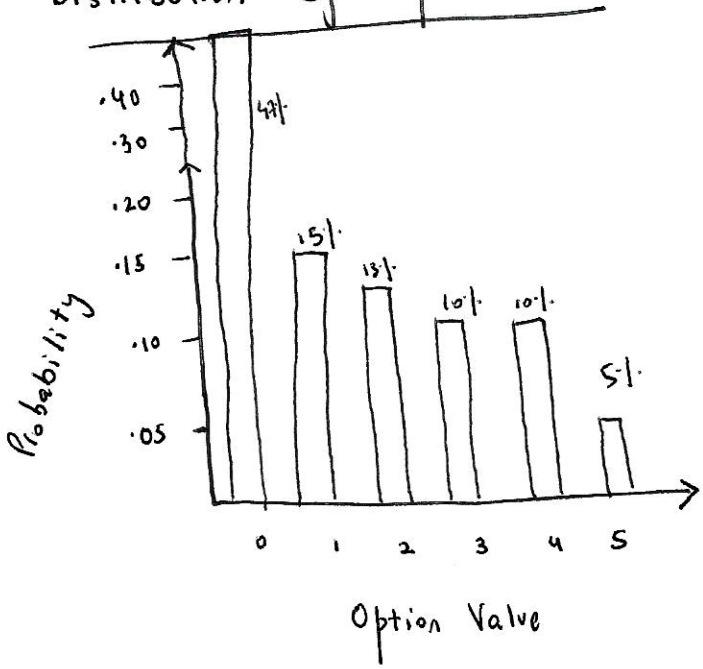
$= .15 + .13 + .10 + .10 + .05 = .53$ or 53%.

5) Mean of Value of Option $= \sum (v) f(x)$

✓ Q5 $\underline{\underline{1.36}}$

~~SD =~~

Variance $= 2.58$ $\qquad$ SD $= \sqrt{\text{Variance}}$

SD $= \underline{\underline{1.60}}$

Distribution of Option Value